Utsav Acharya

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A **Computer Engineering** student with experience in **Machine learning, Backend systems,** and **Applied AI research.** Built and deployed data-driven solutions across **computer vision, LLM systems,** and **forecasting through** research collaborations **and** real-world projects.

WORK EXPERIENCE

Research Intern — 04/2025 – 11/2025
NAAMI

Research Intern working on applied **computer vision** systems in **agricultural and biological monitoring** domains within a physics laboratory setting.

- Developed **deep learning-based disease detection models for rice crops**, improving classification accuracy to ~**93% on curated field datasets** with a **Raspberry Pi-based edge AI application**
- Designed and trained **computer vision pipelines for poultry behavioral monitoring**, enabling automated detection of abnormal movement patterns with **real-time inference capability**.
- Processed and annotated **10,000+ agricultural and livestock image samples** for supervised model training.
- Experimented with **CNN-based architectures (ResNet, EfficientNet variants)** for improved robustness under low-quality field imagery conditions.

Backend Engineer and Data Scientist — 11/2023 – 03/2025
Omdena

Collaborated with **global teams** on applied AI projects focused on **environmental monitoring** and **agricultural analytics**.

- Built ML pipelines for **Combatting Illegal Deforestation in Romania** achieving ~**88% segmentation accuracy in prototype models**.
- Contributed to geospatial analysis workflows for **Mapping Seagras Meadows using remote sensing data and satellite imagery classification techniques**.
- Improved dataset preprocessing efficiency by ~**40%** through optimized ETL pipelines and automation scripts for **Agriculture Analysis on Kenya** .
- Worked in cross-functional teams of **50+ contributors**, coordinating model development, data preprocessing, and evaluation workflows.

ACADEMIC PROJECTS

MEDIWO — LLM-Powered Medical Workflow Optimization

AI-assisted healthcare platform designed to optimize outpatient department (OPD) workflows by automating patient intake, reducing consultation overhead, and generating structured clinical summaries for doctors prior to consultation.

- Engineered a full-stack platform using **React, TypeScript, Tailwind CSS, FastAPI, and MongoDB**, supporting separate patient and doctor dashboards for streamlined clinical interaction.
- Developed an adaptive **AI-driven intake chatbot** that collected and structured patient symptoms, reducing manual intake time by ~**65%**.
- Built a **RAG-based clinical summarization pipeline** using **Llama, LangGraph, and Chroma**, achieving **92% summary relevance accuracy** during internal evaluation.
- Integrated an **OCR-based medical report digitization module**, enabling automated extraction from **500+ uploaded diagnostic documents**.
- Designed a **human-in-the-loop feedback framework** that logged physician corrections as reward signals, improving summary precision by **18% across iterative retraining cycles**.

Compliance Guide — Intelligent Compliance Analysis System

AI-powered compliance intelligence platform built to simplify complex policy documents and automate security framework analysis for faster gap identification and remediation.

- Built a **document intelligence pipeline** using **locally hosted Llama models** for automated policy summarization and contextual compliance interpretation.
- Integrated a **RAG-based retrieval system** that improved compliance recommendation accuracy from **76% to 95%**.
- Developed a **comparative evaluation framework** using dual LLM inference pipelines to benchmark response consistency across compliance scenarios.

- Processed and indexed **1,200+ compliance documents** across security standards including ISO 27001 and SOC 2.
- Reduced average compliance analysis time from **4 hours to under 20 minutes per document set**.

Volt Vision — Household Energy Demand Forecasting for Nepal

Machine learning-based forecasting system for predicting household energy demand and electricity pricing trends in Nepal.

- Processed and analyzed **30,000+ household energy consumption records** using **Pandas and NumPy** for large-scale demand pattern analysis.
- Trained and optimized **XGBoost forecasting models**, achieving **14% lower MAE** than baseline regression approaches.
- Implemented **SARIMAX and ARIMAX models** for multivariate time-series forecasting of household demand and tariff fluctuations.
- Built end-to-end **MLOps deployment pipelines** using **ZenML and MLflow**, enabling experiment tracking and reproducible model versioning.
- Improved forecasting reliability for seasonal demand spikes with **91% prediction accuracy during peak usage periods**.

EDUCATION

Bachelor's in Computer Engineering — 2023 – present
Pulchowk Engineering Campus — Kathmandu, Nepal
Average Percentage: 82.4

High School — 2021 – 2023
Capital College and Research Centre — Kathmandu, Nepal
GPA: 3.91 + 3.73

SKILLS

Languages — Python, C, C++, JavaScript, SQL

Data Analysis — NumPy, Pandas, SciPy, Regex, Tableau

AI Systems — Scikit Learn, TensorFlow, Pytorch, LangChain, LangGraph, RAG, Prompt Engineering

Backend & Deployment — FastAPI, Flask, Docker, AWS, Linux, MLflow

Mathematics — Linear Algebra, Probability & Statistics, Numeical Optimisation with Calculus

CERTIFICATES

- AWS Solutions Architect Associate
- Google Advanced Data Analytics
- IBM Data Science Professional Certificate
- Stanford Machine Learning Specialization
- Deep Learning.AI Deep Learning Specialization
- Tensorflow Developer Professional Certificate

LEADERSHIP & EXTRACURRICULARS

Data Research Council For Students — 11/2022 – 12/2023
Core Member
- Mentored **200+ students** in **intermediate Python**, providing guidance and support to help them improve their skills.
- Managed **GitHub** repositories, **social media** accounts, and **Discord** servers, ensuring that these platforms were up-to-date and effectively engaged with our community.

Together We Learn — 03/2022 – 08/2023
Co founder
- Co-founded and led an organization of **500+** members
- Conducted **5+ webinars** on programming-related topics, sharing knowledge and insights with a broad audience.
- Wrote **10+ newsletters** focused on **data science** and **programming** in general, providing valuable information and updates to **200+ subscribers**.

Kathmandu Valley Leo Club — 01/2023 – 07/2025
Volunteer
- Participated in different volunteering activities from KVLC Pulchowk.